

Foerster et al. (2016)

Learning to Communicate with Deep Multi-Agent Reinforcement Learning

Condensed report: conclusions, gaps, and experiment/implementation considerations

Scope of this report: This is not a final project architecture. It records what this paper establishes, what it does not establish, and the comparisons/implementation checks it directly motivates for our later experiments.

1. What the Paper Is Actually About

The paper studies how fully cooperative agents in partially observable sequential tasks can learn their own communication protocol instead of being given a predefined language. Every agent receives a private observation, takes an environment action, and may send a limited-bandwidth message. All agents optimise the same team return. The central difficulty is that a useful message only receives positive reinforcement if another agent learns to interpret it correctly and then acts correctly, making protocol discovery a hard coordination and credit-assignment problem.

The main algorithmic comparison is RIAL vs DIAL, but the paper also studies parameter sharing, recurrent architecture, channel noise/discretisation, and no-communication/oracle baselines.

2. RIAL vs DIAL - Core Difference

Aspect	RIAL	DIAL
Communication during training	Discrete communication action learned with Q-learning.	Real-valued message is a differentiable connection between sender and receiver.
Sender feedback	Indirect: sender learns from eventual TD/reward consequences.	Direct: receiver-side loss gradient can flow back through the message to the sender.
Execution	Discrete limited-bandwidth messages.	Also discrete; continuous training activation is thresholded/binarised at execution.
Main advantage	Conceptually simple and genuinely action-like communication.	Richer credit signal; easier discovery of coordinated protocols, especially with delayed/stochastic rewards.
Main cost/assumption	Hard exploration and sender-receiver coordination.	Requires centralised differentiable training and a train/execution message relaxation.

RIAL also factorises environment-action and message Q-values rather than learning a full joint $|U| \times |M|$ action table. DIAL replaces the message Q-output with a message activation. Because DIAL receives a vector-valued gradient through the channel, the paper argues that its learning signal for message content is much richer than the scalar/selected-action TD signal available to RIAL.

Important DIAL detail: The Discretise/Regularise Unit (DRU) adds Gaussian noise and applies a logistic squashing function during training; at decentralised execution the outgoing activation is thresholded into binary messages. The noise is not mainly a model of real wireless packet loss - it is a regulariser that prevents the continuous training channel from hiding too much information and reduces discretisation error.

3. What the Paper Compared

- RIAL with shared parameters across agents.
- RIAL-NS: RIAL without parameter sharing (independent agent networks).
- DIAL with shared parameters across agents.
- DIAL-NS: DIAL without parameter sharing.
- NoComm: no-communication baseline with shared parameters.
- Oracle: access to true state; used as the upper reference for normalising reported rewards.

This means the paper is not only asking "RIAL or DIAL?" It also asks whether centralised learning through shared parameters helps, whether differentiable communication adds value beyond sharing parameters, and whether the learned communication is actually necessary relative to NoComm.

4. Experimental Findings

4.1 Switch Riddle

The switch acts as a one-bit communication/memory channel. An agent must eventually announce that all agents have visited the interrogation room; premature declaration gives -1, correct declaration gives +1, otherwise reward is zero. The environment therefore tests whether agents can invent a protocol that stores distributed historical information in one bit.

- With 3 agents, all four communication variants can learn an optimal policy, but DIAL with parameter sharing reaches optimum substantially faster than RIAL.
- Parameter sharing speeds both methods.
- With 4 agents, DIAL with parameter sharing is best; RIAL without parameter sharing cannot beat NoComm.
- DIAL without sharing performs roughly similarly to shared RIAL in the harder setting, supporting the authors' claim that the differentiable message gradient is a richer/robuster learning signal.
- The authors extract and inspect the learned 3-agent DIAL protocol and show that it corresponds to an optimal strategy - important evidence that the message really carries task-relevant state/history.

4.2 Colour-Digit MNIST

Two agents observe coloured MNIST digits but the colour label and digit value are hidden. Only one bit can be communicated. The reward structure makes different latent encodings possible, with one encoding (digit parity) more valuable. This tests whether agents can coordinate on which information deserves the limited channel.

- DIAL substantially outperforms the other approaches.

- RIAL tends to fluctuate around local minima in protocol space rather than reliably finding the optimal protocol.
- NoComm remains at the zero baseline because the missing private information is genuinely necessary.
- The appendix explains why RIAL can struggle under this antisymmetric/stochastic reward structure: expected TD signals for the sender can cancel before a protocol exists, whereas DIAL's receiver-conditioned gradient need not cancel.

4.3 Multi-Step MNIST

Agents exchange one bit per step across multiple time steps and must ultimately identify a digit. The task forces messages to form a temporal code: information sent early must remain useful many steps later.

- DIAL reaches near-optimal/optimal performance; RIAL fails to learn the task in the reported setting.
- This is the strongest demonstration of DIAL's credit-assignment advantage: gradients can optimise an early message with respect to rewards and errors occurring several time steps later.
- The authors extract the bit patterns and show that agents learned a binary encoding/decoding scheme for digits rather than merely exploiting an unrelated shortcut.

4.4 Effect of Channel Noise / DRU

- Without training noise, DIAL learns centred continuous activations that are poor candidates for later binarisation.
- With noise, activations separate into two modes, making the learned representation robust to discrete execution.
- The authors state that adding noise was essential for successful training in their experiments.
- The appendix further analyses how noise limits effective channel capacity: larger noise reduces how many reliably distinguishable values the continuous bottleneck can encode.

5. Main Conclusions Supported by the Paper

1. Communication protocols can emerge from cooperative task reward without manually assigning semantics to messages.
2. Differentiable inter-agent communication can make protocol discovery substantially easier than treating messages only as discrete RL actions, especially when rewards are delayed or stochastic.
3. Centralised learning matters in two distinct ways: parameter sharing reduces the learning burden, while DIAL additionally allows gradients to cross agent boundaries.
4. Parameter sharing is a major experimental factor, not a minor implementation detail. Shared networks can still specialise because agents receive different observations/hidden states and an agent-ID input.
5. Recurrent memory is important because communication protocols map histories to message sequences; the agents must remember past observations/messages/actions.
6. Limited-bandwidth channels can still support useful, structured protocols. The paper demonstrates one-bit communication and multi-step binary codes.
7. Protocol analysis is necessary. The paper does not rely only on return curves; it extracts learned decision rules/codes to show what the communication is doing.
8. The training communication channel must be designed with deployment discretisation in mind. DIAL's noise + logistic bottleneck is a concrete example.

6. Important Methodological Details (Paper-Specific)

- RIAL and DIAL use the same basic per-agent recurrent architecture: observation/task embedding + previous message + previous action + agent ID, followed by a two-layer GRU-based recurrent network and output MLP.

- The paper reports 128-dimensional embeddings/hidden layers, batch normalisation on incoming messages, ReLU activations, and GRUs.
- Default experimental settings: epsilon-greedy epsilon = 0.05, discount gamma = 1, target network reset every 100 episodes, batches of 32 parallel episodes, RMSProp momentum 0.95, learning rate 5e-4.
- DIAL uses DRU noise sigma = 2 by default; Multi-Step MNIST reports sigma = 0.5.
- These values are reproducibility references from this 2016 DQN setup, not parameters we should automatically copy into a PPO/MAPPO implementation.

7. Gaps / Limitations to Keep in Mind

The paper itself explicitly presents the work as an early step rather than a complete theory of learned language. The following limitations are the most relevant when using it as a foundation:

Centralised-training assumption: DIAL needs a differentiable graph across agents during training. This is reasonable in simulation but is privileged relative to fully decentralised learning.

Continuous-training / discrete-execution mismatch: DIAL trains through real-valued activations and discretises at execution. DRU reduces, but does not conceptually remove, this mismatch.

Small, controlled tasks: Switch and MNIST isolate communication learning well, but do not test rich embodied navigation, physical uncertainty, task allocation, or large dynamic environments.

Small team scale: The showcased experiments involve very few agents; scalability to large teams is not established.

Homogeneity and parameter sharing: The strong benefit of shared parameters is easiest to exploit with similar agents. Heterogeneous roles/action spaces require additional design decisions.

No realistic networking model: There is no study of packet loss, latency, interference, range, contention, or energy/bandwidth cost as real communication phenomena.

Communication availability is largely fixed: The paper is focused mainly on learning message content, not explicitly learning whether to communicate or whom to target.

Interpretability is easy only because tasks are simple: The authors can decode protocols in these benchmarks, but richer tasks may produce useful yet opaque latent conventions.

8. Experiments and Comparisons This Paper Directly Motivates for Us

Boundary: These are conclusions/controls suggested by this paper. They are not a final project roadmap; later papers may change the architecture or add stronger baselines.

- **NoComm baseline:** Train the same task without communication. If communication methods do not beat this, either the task does not require information exchange or the communication mechanism is not learning.
- **RIAL-style vs DIAL-style communication:** Initially compare discrete RL-learned messages against differentiable training-time messages. This tests whether the same credit-assignment difference appears in our environment.
- **Start under an ideal channel:** First test with zero packet loss/latency and no explicit communication cost. We should prove that protocol learning works before adding networking failure modes.
- **Parameter-sharing ablation:** Where architecture permits it, compare shared vs non-shared parameters or at least document exactly which modules are shared. Foerster shows this can change the result dramatically.

- **Message-capacity control:** Keep communication deliberately bottlenecked and vary message size/bits. Otherwise agents may simply transmit a large latent state and the "communication protocol" becomes hard to interpret.
- **Protocol-semantic analysis:** Measure how messages correlate with private observations/events and how receiving them changes downstream actions. Do not report only reward.
- **Communication ablation at evaluation:** Disable, randomise, or scramble messages after training. A significant performance drop is evidence that agents genuinely depend on the learned protocol.
- **DIAL discretisation/noise ablation:** If we implement DIAL, compare the DRU/noise settings and check train-vs-execution performance. The paper shows noise is an important engineering factor, not optional decoration.
- **Learning efficiency and stability:** Compare not only final success/return but convergence speed, variance across seeds, and failure/local-minimum behaviour. Much of DIAL's value in the paper is learning efficiency and reliability.
- **Later channel-stress tests:** After the ideal-channel baseline works, packet loss, latency, bandwidth/cost constraints can be layered on as separate experiments. These are natural extensions precisely because Foerster does not test them as realistic network effects.

9. Bottom Line from Foerster et al.

What we should carry forward: emergent communication needs partial observability, a real information bottleneck, and careful credit assignment. DIAL demonstrates that cross-agent gradients can make protocol learning much easier than pure trial-and-error message actions. The paper also makes parameter sharing, recurrent memory, no-communication controls, discretisation/noise, and protocol inspection explicit experimental factors.

What this paper does NOT decide for us: the final MARL backbone, the final communication architecture, whether RIAL or DIAL should remain our main method, whether communication should be gated/targeted, or which realistic channel constraints should define the final study. Those decisions should be made only after combining this paper with the later literature.

Source

J. N. Foerster, Y. M. Assael, N. de Freitas, and S. Whiteson, "Learning to Communicate with Deep Multi-Agent Reinforcement Learning," arXiv:1605.06676, 2016.

Paper: <https://arxiv.org/pdf/1605.06676>